A065 Triadic–Casimir / PKLSA
Blind Falsifier v0.3.2 – Convergence and Independent Provider Tranche

SST-Workbench; self-contained computational report

Run: A065-v0.3.2___20261014T145125153813Z-9b04106065

Scientific status: This is a geometry and synthetic numerical pilot. Neither the time-dependent incompressible Euler equations nor a microscopic metal response nor a measured Casimir force has been certified. All preregistered failures are retained.

1 Identification and provenance

Run mode	QUALIFY
Framework	NOT_RUN
Blind campaign	SYNTHETIC_GATES_FAILED
PKLSA source points	300 KnotPlot trefoil (3_1)
Original source SHA-256	`a60b1a95a00110e3cb32087e...`

2 The three response channels

The June three-observable interpretation is retained as three distinct measurements, not as the sum of three forces:

$$\mathcal{O}[X;a] = \begin{pmatrix} \mathbf{a}_{\mathrm{bulk}}[X;a] \\ \Delta\nu/\nu[X;a] \\ P_{\mathrm{boundary}}[X;a] \end{pmatrix}.$$

With constant-density incompressible Euler kinematics,

$$\nabla\cdot\mathbf{u} = 0, \qquad \boldsymbol{\omega} = \nabla\times\mathbf{u}, \tag{1}$$

$$\nabla^2 p = \rho_M\left(\tfrac{1}{2}|\boldsymbol{\omega}|^2 - \|S\|_F^2\right), \qquad S_{ij} = \tfrac{1}{2}(\partial_i u_j + \partial_j u_i). \tag{2}$$

In a *stationary, irrotational* region the conditional Swirl-Clock reduction gives

$$p + \tfrac{1}{2}\rho_M|\mathbf{u}|^2 = C, \qquad N^2 = 1 - |\mathbf{u}|^2/c_\star^2 \implies \nabla\ln N = \frac{\nabla p}{\rho_M c_\star^2 N^2}.$$

This identity is conditional on selecting the same effective velocity in Bernoulli and the clock hypothesis; it is *not* a proof of gravitation.

3 Gauge-invariant clock difference and two-surface toy

The proposed coupling requires an as-yet-unverified gauge transformation of the atomic phases:

$$\Theta_{AB} = \theta_B - \theta_A - \sum_{e\in\mathcal{P}}\vartheta_e, \tag{3}$$

$$\dot\Theta_{AB} = \Omega_B^{(0)}N_B - \Omega_A^{(0)}N_A + \sum_{e\in\mathcal{P}}E_e. \tag{4}$$

In the *classical scalar test only* the two-sheet free-energy density is evaluated numerically from

$$\frac{\Delta\mathcal{F}}{A} = \frac{T}{4\pi}\int_0^\infty k\,\ln[1 - r_1 r_2\exp(-2ka)]\,dk.$$

The toy reflectivities r_j are model inputs and are not deduced from conducting metal atoms. There is no claimed quantum zero-point normalization.

4 PKLSA geometry and dynamical resolution

The source is a real $N = 300$ centerline, but the filament model is a reduced regularized ODE with RK4, not a 3D Euler PDE. All three spatial grids use the same original source, the same $\epsilon = 0.12$, $\Gamma = 1$, and $T = 0.04$. No grid-dependent core fitting is permitted. Its dimensionless velocity kernel is

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \sum_j \frac{\Delta \mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+\frac{1}{2}})}{\left(|\mathbf{x} - \mathbf{X}_{j+\frac{1}{2}}|^2 + \epsilon^2\right)^{3/2}}.$$

Quantity	Registered value
Final time T	0.04
Time-step relative discrepancy	2.0246251e-11
Spatial error (75 vs 150)	0.087289647
Spatial error (150 vs 300)	0.03266325
Spatial error (75 vs 300)	0.12122164
Predeclared spatial threshold	0.05
Historical 75–150 gate	**FAIL**

5 Independent source provider reconciliation

A single pre-run source manifest is verified by both the portable and official framework pipelines; each must recheck the exact raw bytes. A raw provider check is *geometry evidence only*, never an independent 3D Euler or Casimir-force certification.

Provider status	`PASS_RAW_HASH_AND_XYZ_PARSER_ONLY`
Provider SHA-256	`74cc9349324a15f4c1d4ca1a`
Provider points	118

6 Independently sealed datasets

The datasets use separate hash manifests. Different discretizations of the same original trefoil are *not* independent upstream providers.

Dataset	Seal digest (prefix)
D01_EULER	e79a4e0cfc370eb08a05d0288e8138a8...
D02_CLOCK_LINK	60729d7f5a206fdc025dbbde95d9284c...
D03_TWO_SURFACE	7d5e24b065ee9d6cbd43e3a9ce9a6d08...

7 Gate ledger

Gate	Status	Metric / reason
`D01_G01_EULER_DIV`	PASS	0.0
`D01_G02_POISSON`	PASS	0.0
`D01_G03_POTENTIAL_NULL`	PASS	0
`D01_G04_IRROTATIONAL_CLOCK_VARIATION`	PASS	0.0008849583697424501
`D01_G07_FILAMENT_RK4_DT_HALVING`	PASS	2.0246251170010892e-11
`D01_G08_FILAMENT_SPATIAL_REFINEMENT`	FAIL	0.08728964728469735
`D01_G09_FILAMENT_SPATIAL_150_300`	PASS	0.03266324959089548
`D01_G10_FILAMENT_SPATIAL_75_300`	FAIL	0.12122163649103088
`D01_G05_PKLSA_SOURCE_INTEGRITY`	PASS	300
`D01_G06_PKLSA_FINITE_BIOTSAVART`	PASS	0.002346617627425827
`D02_G01_GAUGE`	PASS	5.551115123125783e-17
`D02_G02_PHASE_RATE`	PASS	5.551115123125783e-17
`D02_G03_HOLONOMY`	PASS	2.0816681711721685e-17
`D02_G04_RESPONSE_POSITIVE`	PASS	0.0
`D02_G05_SHEET_STABLE`	PASS	1.6305756807341665
`D02_G06_NO_RESPONSE_CONTROL`	PASS	0.0
`D03_G01_ENERGY_STRESS`	PASS	1.141916836020315e-09
`D03_G02_DECOUPLED_NULL`	PASS	0.0
`D03_G03_FINITE_TOY_RESPONSE`	PASS	negative=28, total=28

Non-pass gates: D01_G08_FILAMENT_SPATIAL_REFINEMENT, D01_G10_FILAMENT_SPATIAL_75_300.

8 Two-surface synthetic fit

Pair	T	Fitted exponent
A/B	0.8	2.198691
A/B	1.6	2.198691
B/B	0.8	2.1234009
B/B	1.6	2.1234009

9 Qualification boundaries and reproducibility

- E010/Fremlin evidence is checked using one identical pre-run manifest in portable and framework mode. Verified raw geometry is not physical independent replication. No invented replacement is accepted.
- No calibrated moving metal surface, atom–link response, normal-mode quantum statistics, independently evaluated Casimir stress, or gravity-equivalence test is available.
- A065 physical discovery G3 is **UNRESOLVED / NOT RUN**; a successful program run cannot promote it to PASS.
- The frozen SST Falsifier Framework v1.0.7 remains external, unmodified, and authoritative when present. Its own TeX report and frozen gate ledger are kept in the per-run framework instance.
- This `.tex` contains **no** `\input`, `\include`, bibliography file, figures, or generated font dependencies. Recompile using `pdflatex` twice in the same folder.
- Each execution receives a UTC-stamped and random-suffixed run ID. Export files are opened create-only; previous experiments are never deleted or overwritten.

10 v0.3.2: Fixed-core spatial convergence and independent simulation provider

The historical 75/150 failure remains registered. The 600-node geometry is a periodic cubic interpolation of the same 300 PKLSA points, not independent source data.

$$X_{2j+1} = \frac{-X_{j-1} + 9X_j + 9X_{j+1} - X_{j+2}}{16}, \qquad X_{2j} = X_j.$$

For all spatial comparisons the original $T = 0.04$, $\Gamma = 1$, and $\epsilon = 0.12$ remain unchanged. The error statistic is the RMS difference of final node positions divided by the coarse-grid RMS displacement.

Pair	Relative error	Gate at 5%
75_vs_150	0.08728965	FAIL
150_vs_300	0.03266325	PASS
300_vs_600	0.008491986	PASS
75_vs_300	0.1212216	FAIL

Time refinement: $N = 300$, $4 \to 8$ steps: 1.691463e-09; $N = 600$, $8 \to 16$ steps: 8.115306e-11.

Independent Fremlin lane: SIMULATION_NUMERICAL_FAIL. Source label alone is not accepted: raw SHA-256, a separate C++17 executable, and spatial and temporal controls must all be verified. Missing raw geometry means unresolved, not a simulated pass.

Physical scope: these are two independent upstream geometries and independently implemented filament-ODE numerics only if Fremlin passes. They do not certify a second full Euler/PDE dynamics provider, Casimir stress, or gravitation.